

Ethanol Blending in Gasoline - Mexico

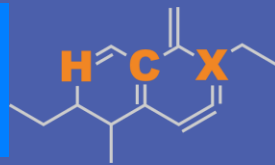
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20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

FARO90



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+52 55 6122 2255

Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Mexico



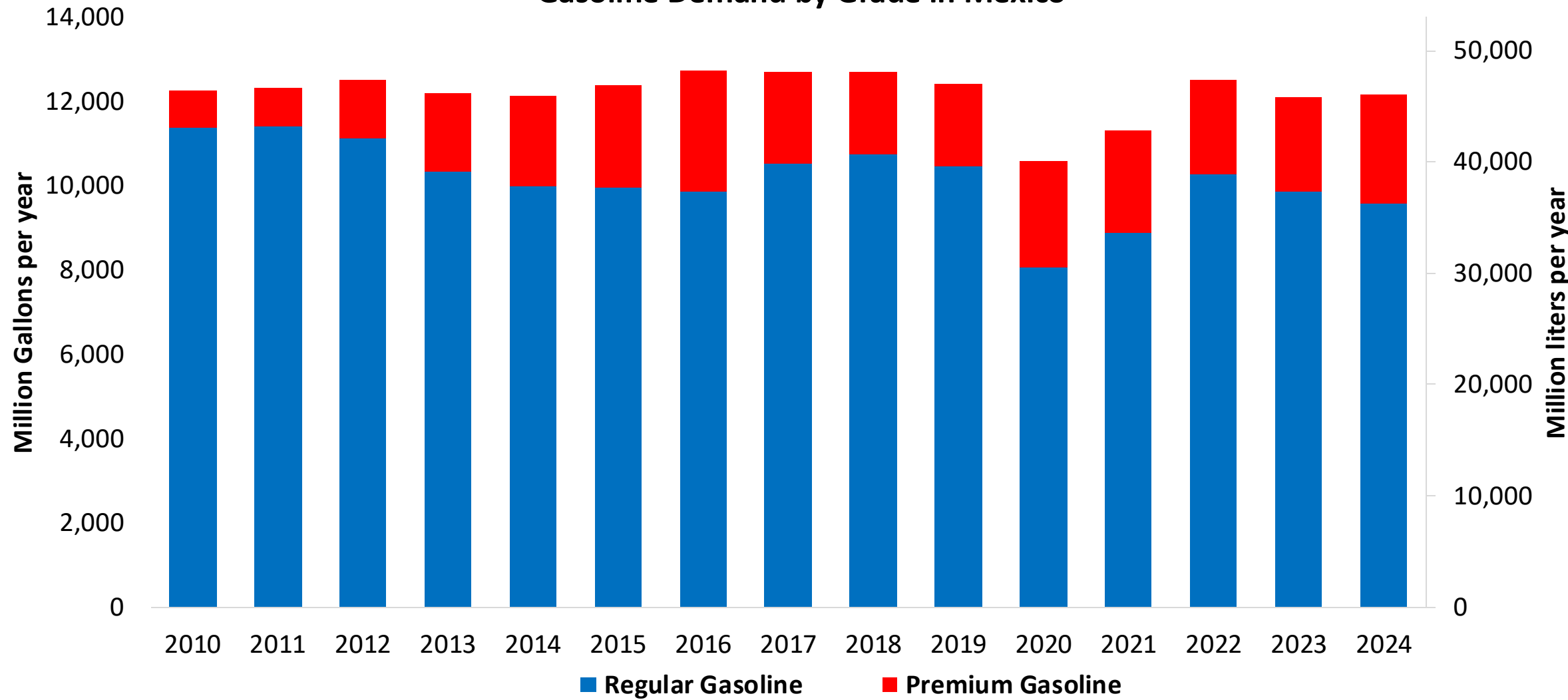
In 2024, gasoline consumption was 12,200 million gallons (46,000 million liters). There are two grades of gasoline, Regular (AKI 87) and Premium (AKI 91). Regular gasoline is the main grade in the market, representing 90% of the volume. There are stringent gasoline specifications for major metropolitan areas in Mexico City, Guadalajara and Monterrey. Demand is much higher than production, making necessary to import 40% of gasoline mainly from the United States.

Gasoline quality norm (NOM-016-CRE de 2016) allows blends with a maximum of 5.8% v/v of ethanol with the exception of metropolitan areas of Valle de Mexico, Guadalajara and Monterrey. Current ethanol use is equivalent to an average of 1.06% in the Rest of the Country grade gasoline. In 2025, a new biofuels law was approved that fosters the production with agricultural and other organic residuals. Recently there are new state level efforts to increase ethanol consumption (e.g. Tamaulipas).

Source: SIE - SENER, 2024

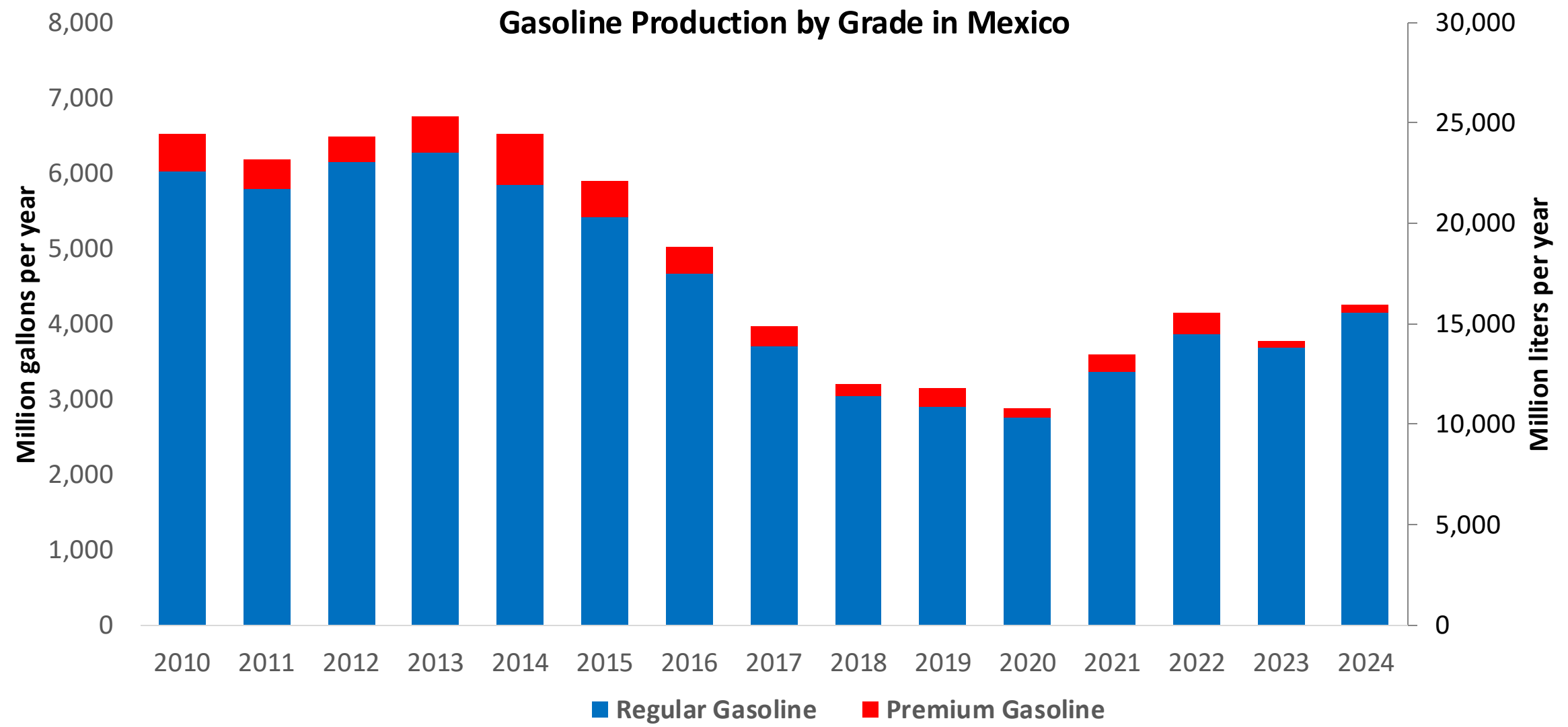
Gasoline Demand in Mexico

Gasoline Demand by Grade in Mexico



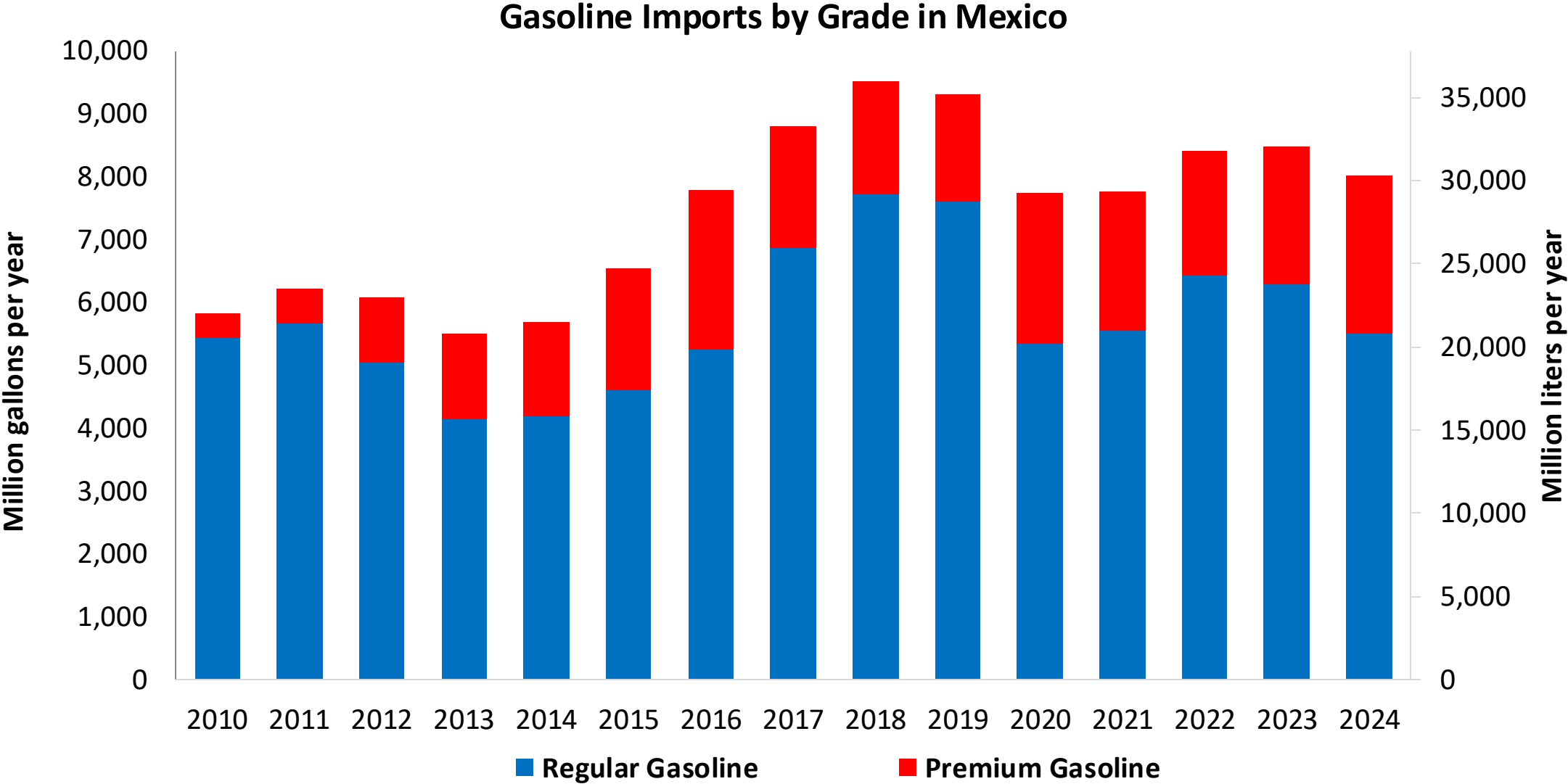
Source: SIE - SENER, 2025

Gasoline Production in Mexico



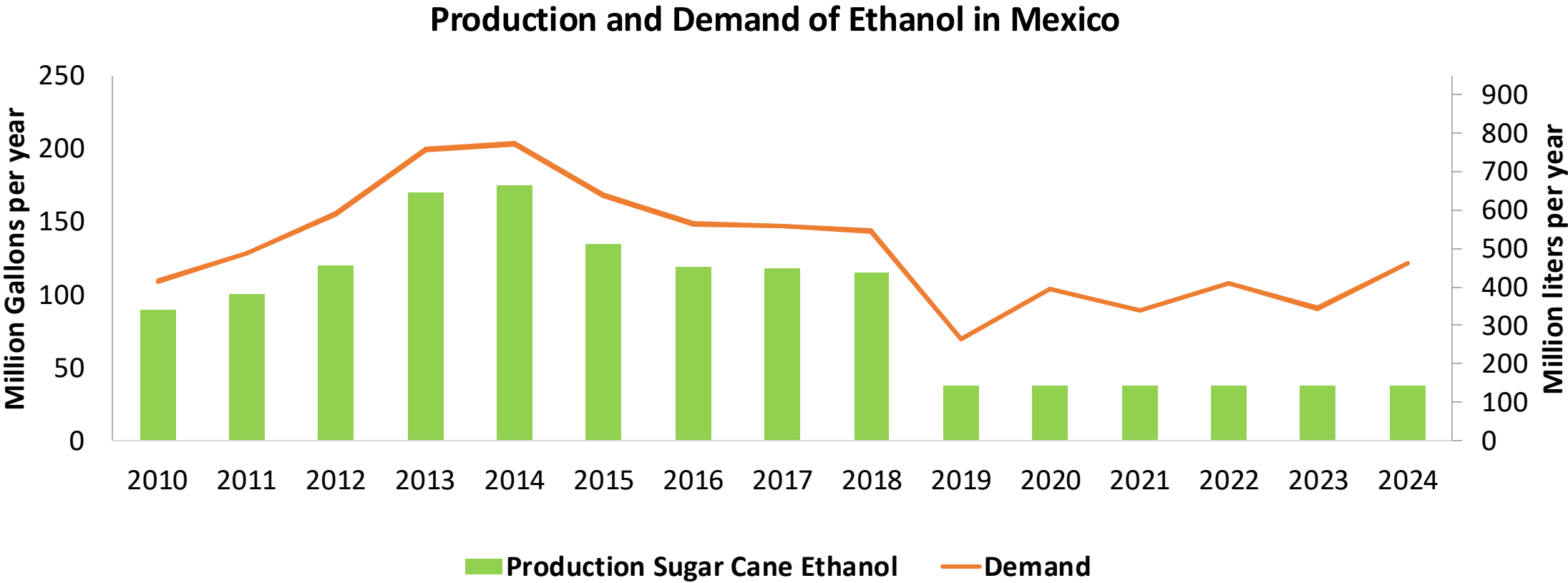
Source: SE - SENER, 2025

Gasoline Imports in Mexico



Source: SE - SENER, 2025

Ethanol Balance in Mexico



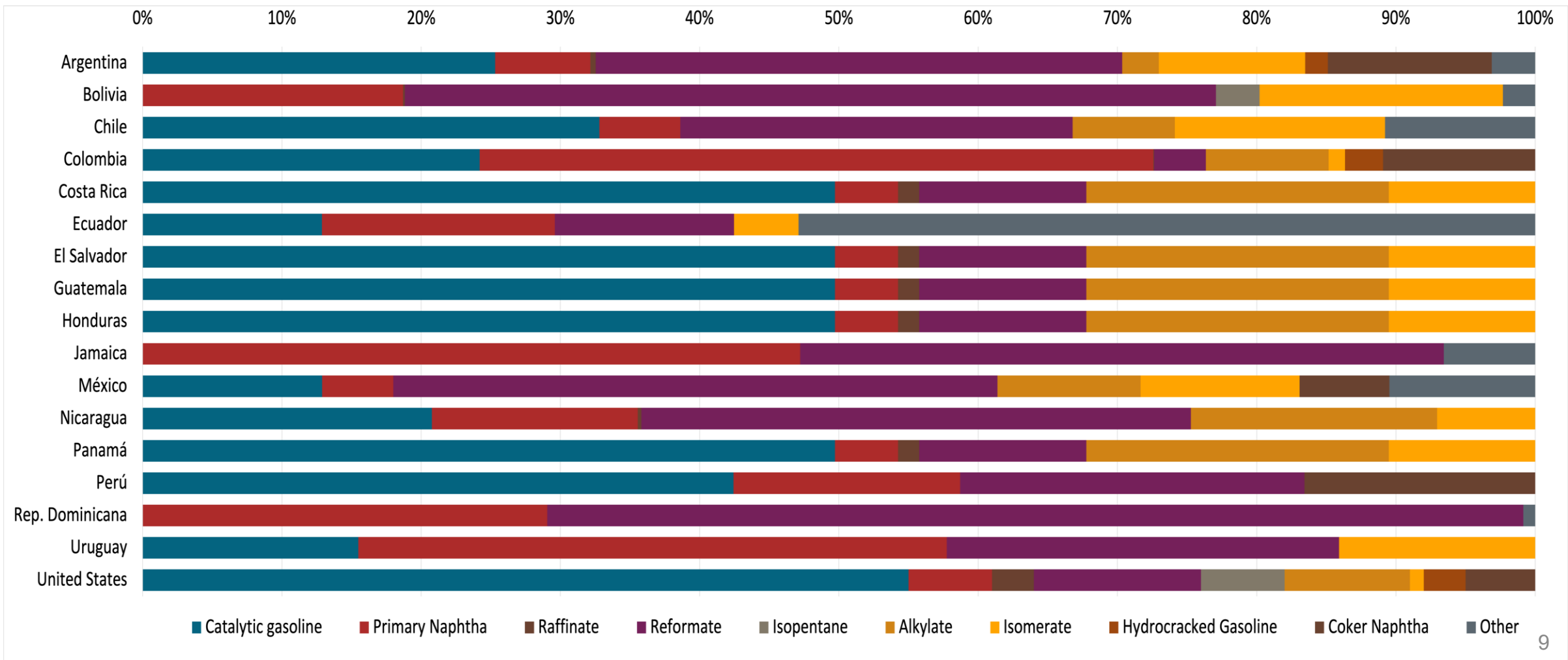
Source: CEDRSSA, 2020; EIA, 2025; Sanchez, 2014.

Gasoline Quality in Mexico

Name	NOM-016-CRE-2016						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2016						2017			
Applicability	Mexico City	Mexico City	Guadalajara and Monterrey	Guadalajara and Monterrey	Rest of the country	Rest of the country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 2 %v/v	< 2 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 25 %v/v	< 25 %v/v	< 32 %v/v	22,6% v/v	Report	22,6% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 10 %v/v	< 10 %v/v	< 11,9 %v/v	< 11,9 %v/v	Report	< 2,5 mg/l	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	-	-	-	-	-	-	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON		94		94		94	> 95	> 95	> 98	> 98
MON	82		82		82		> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m
Ethanol (EtOH)	Not allowed	Not allowed	Not allowed	Not allowed	> 5,8 %v/v	> 5,8 %v/v	< 5 %v/v	< 10 %v/v	< 5 %v/v	< 10 %v/v
RVP 37.8°C (Summer)	< 62 kPa	< 62 kPa	< 62 kPa	< 62 kPa	< 69 kPa North, < 62 kPa Center, South, Pacific	< 69 kPa North, < 62 kPa Center, South, Pacific	< 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa				
RVP 37.8°C (Transition)	< 54 kPa	< 54 kPa	< 54 kPa Guadalajara	< 54 kPa Guadalajara						
MTBE							-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	-	-	Based on oxygen content	< 22 %v/v	Based on oxygen content	< 22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



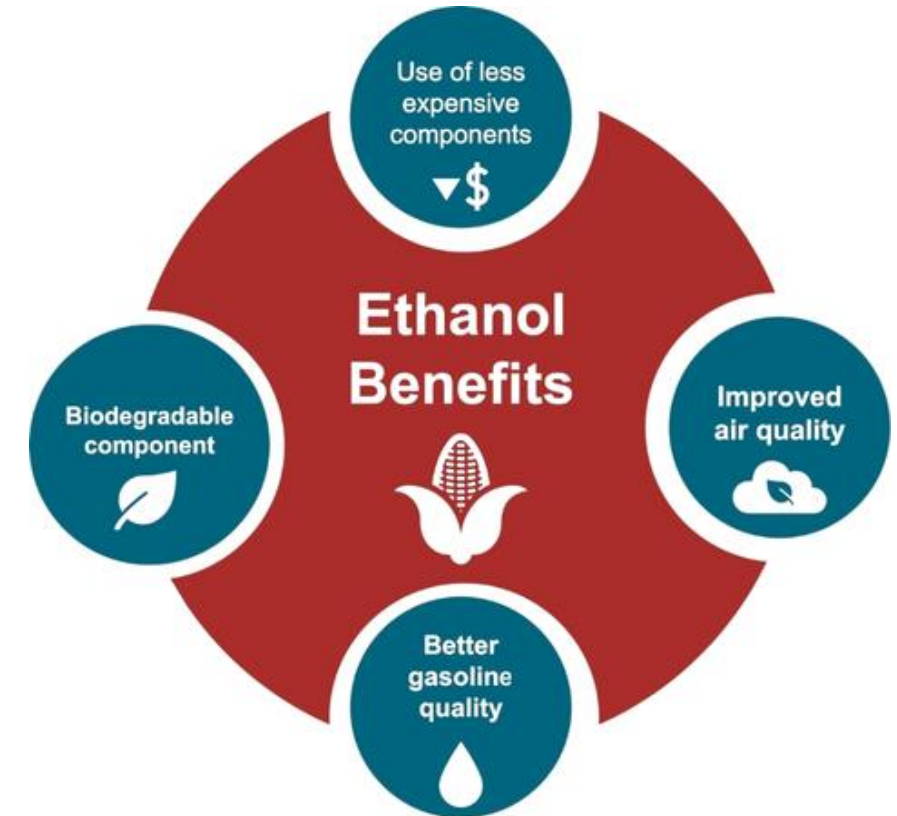
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

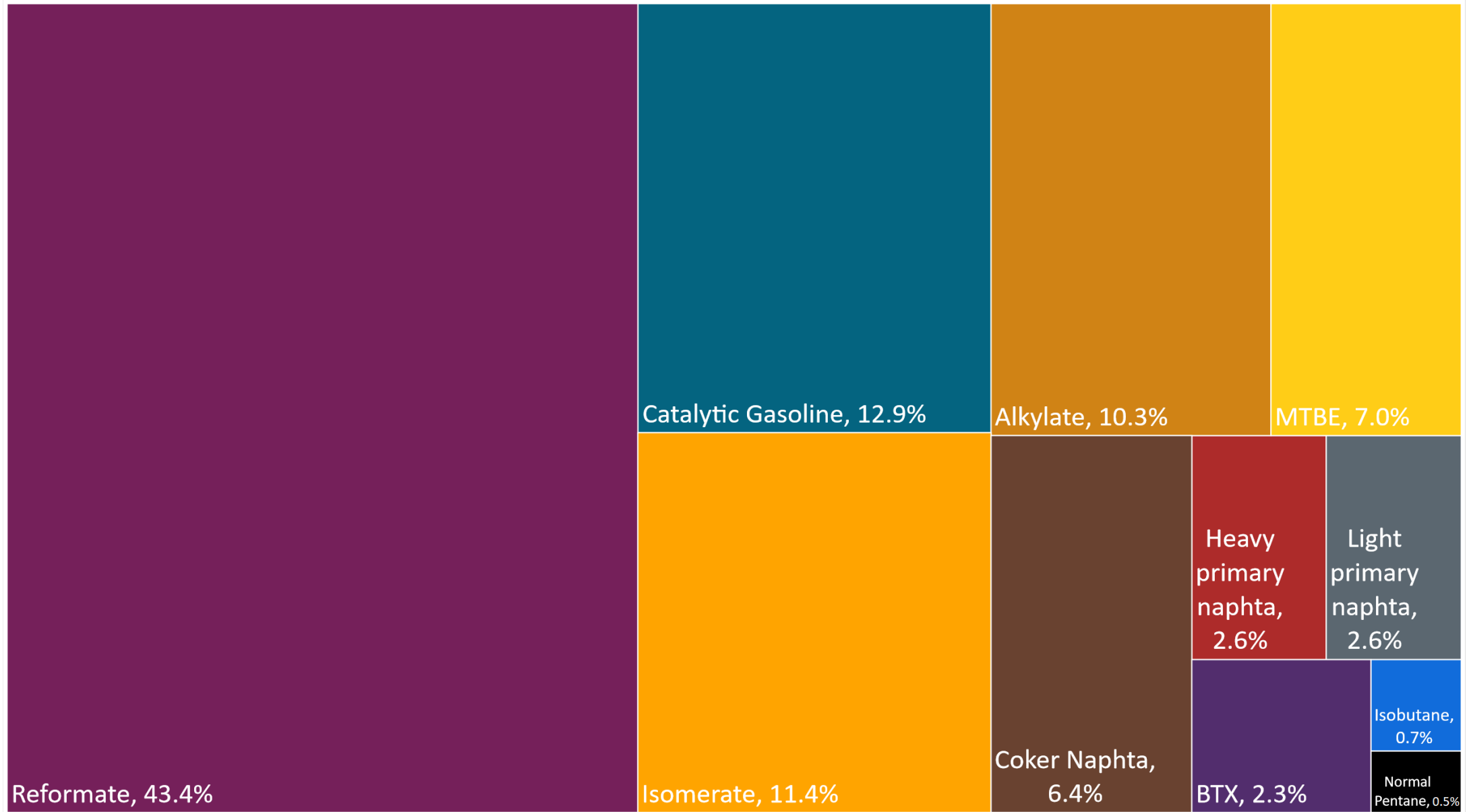
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

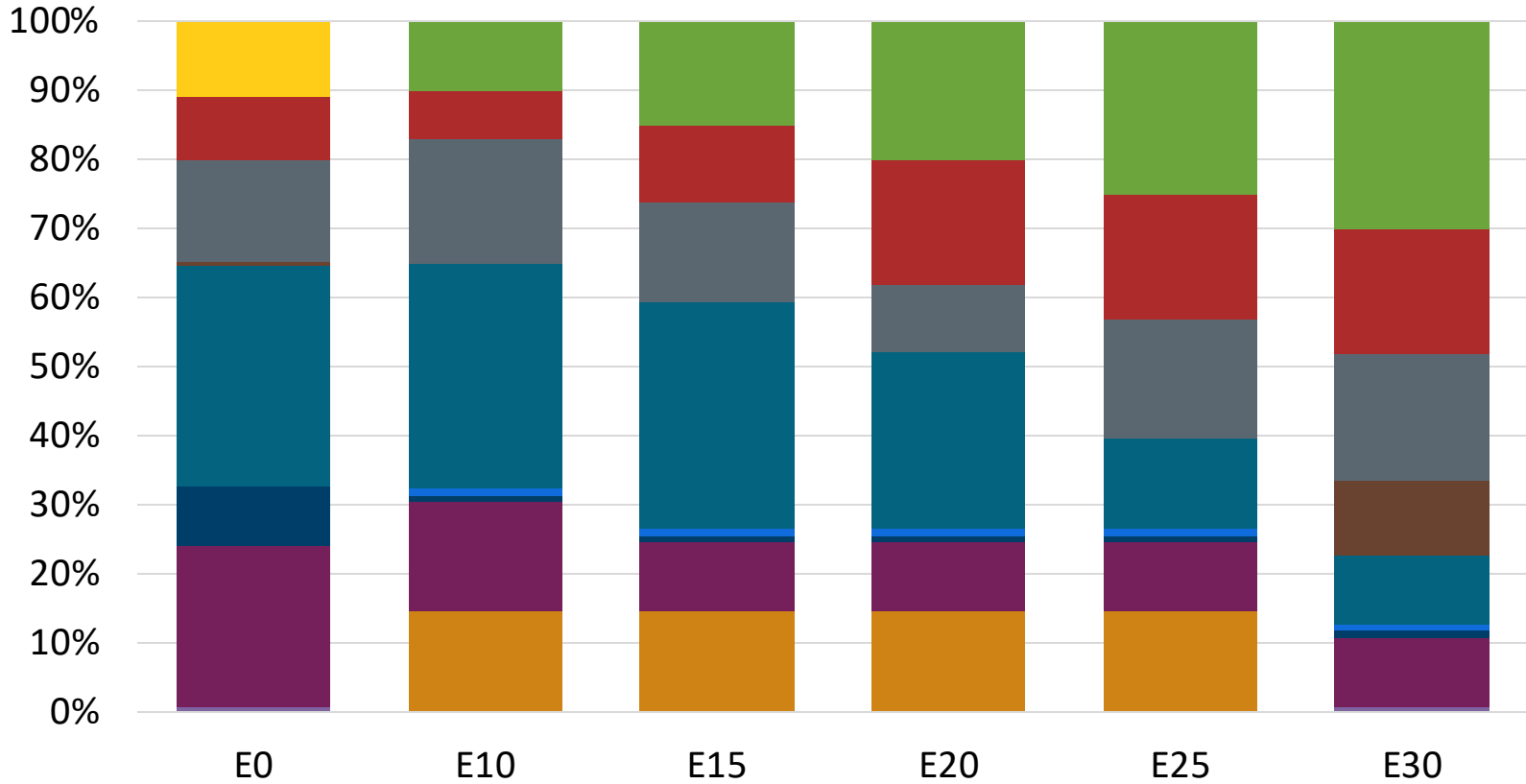
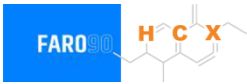
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components

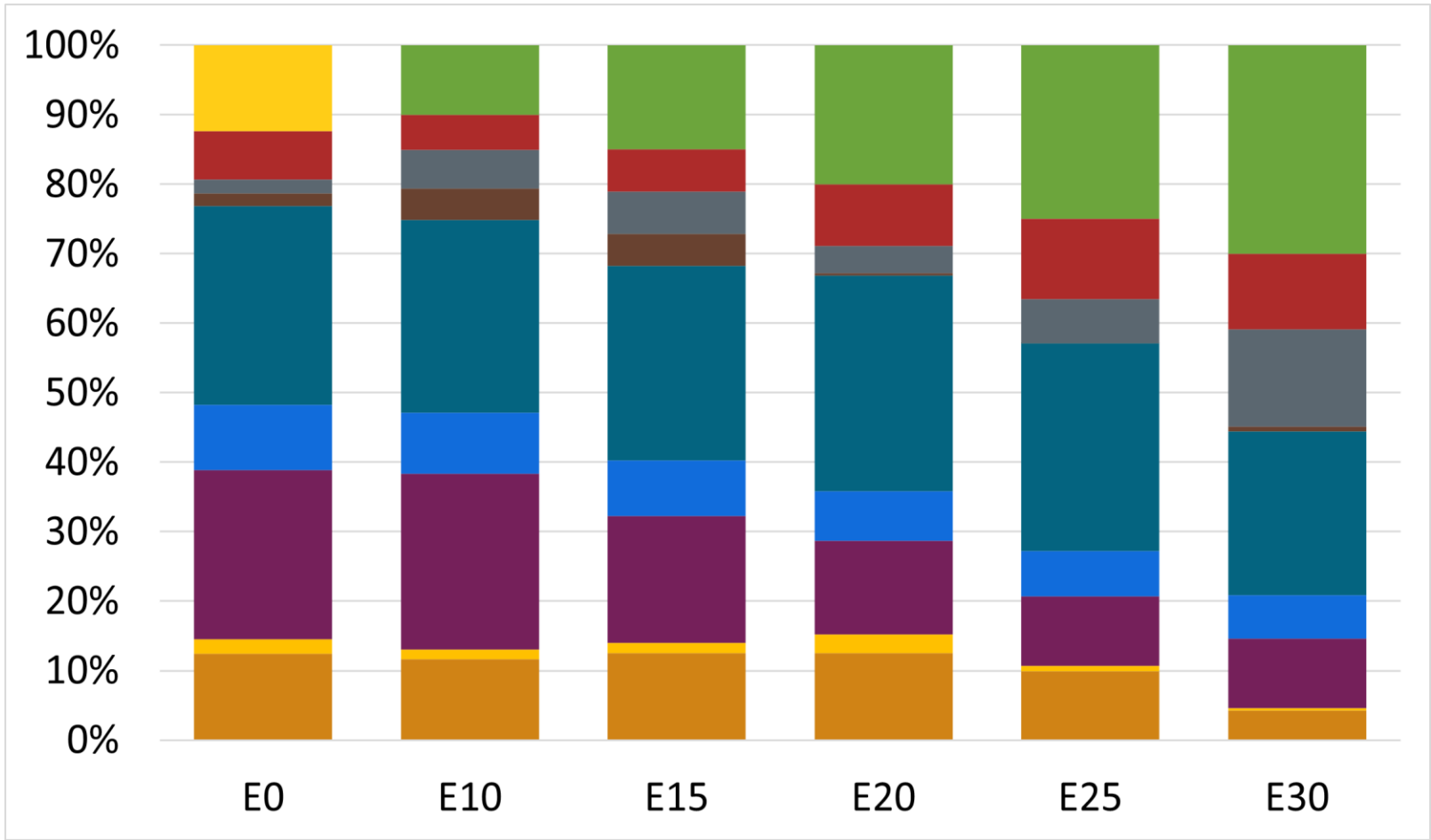


Regular Gasoline RP – Constant Octane



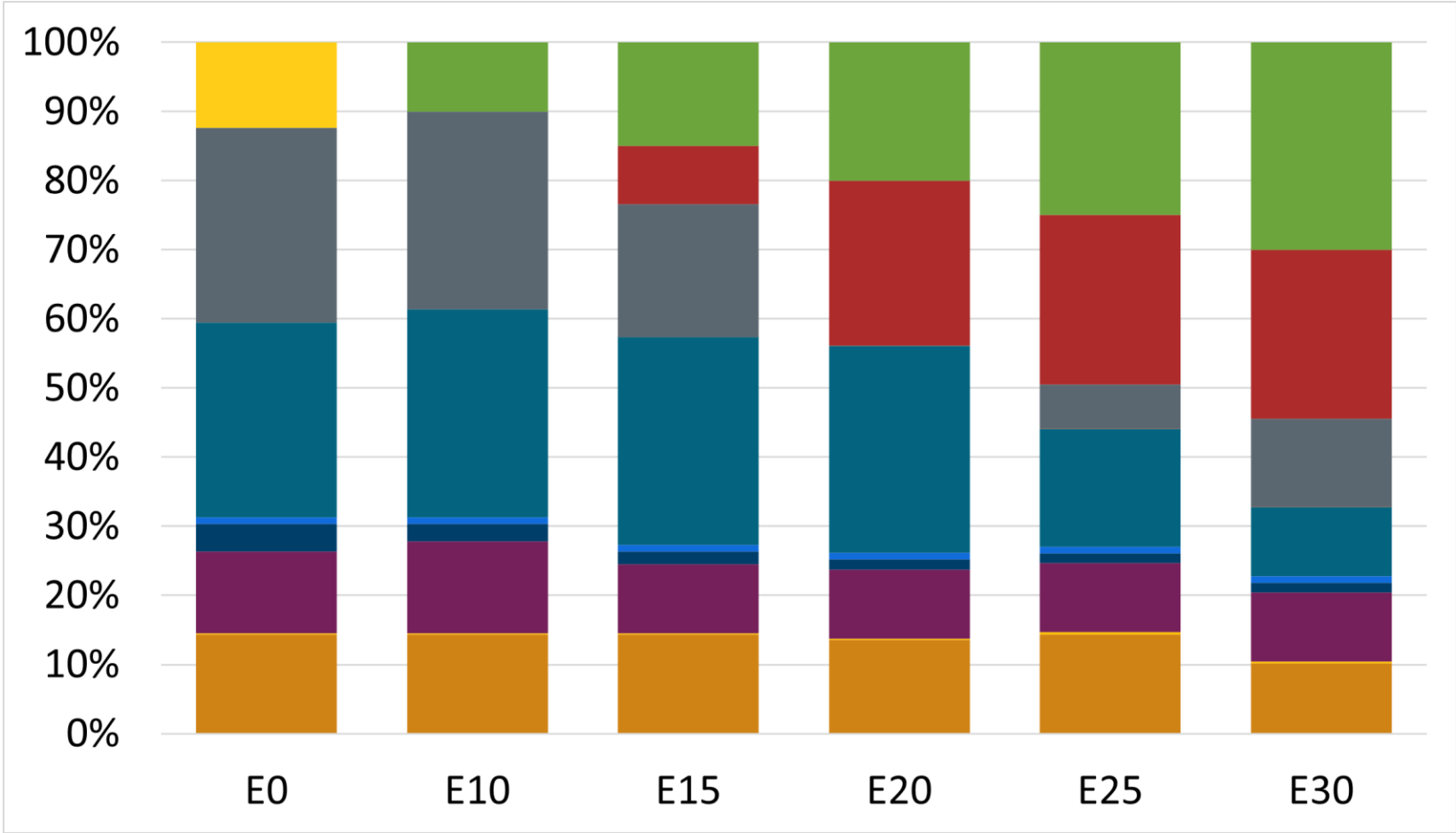
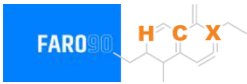
Octane (AKI)	87.1	87.0	87.0	87.0	87.0	87.0
Price (US\$/gal)	\$ 2.245	\$ 2.286	\$ 2.210	\$ 2.138	\$ 2.067	\$ 2.005

Premium Gasoline RP – Constant Octane



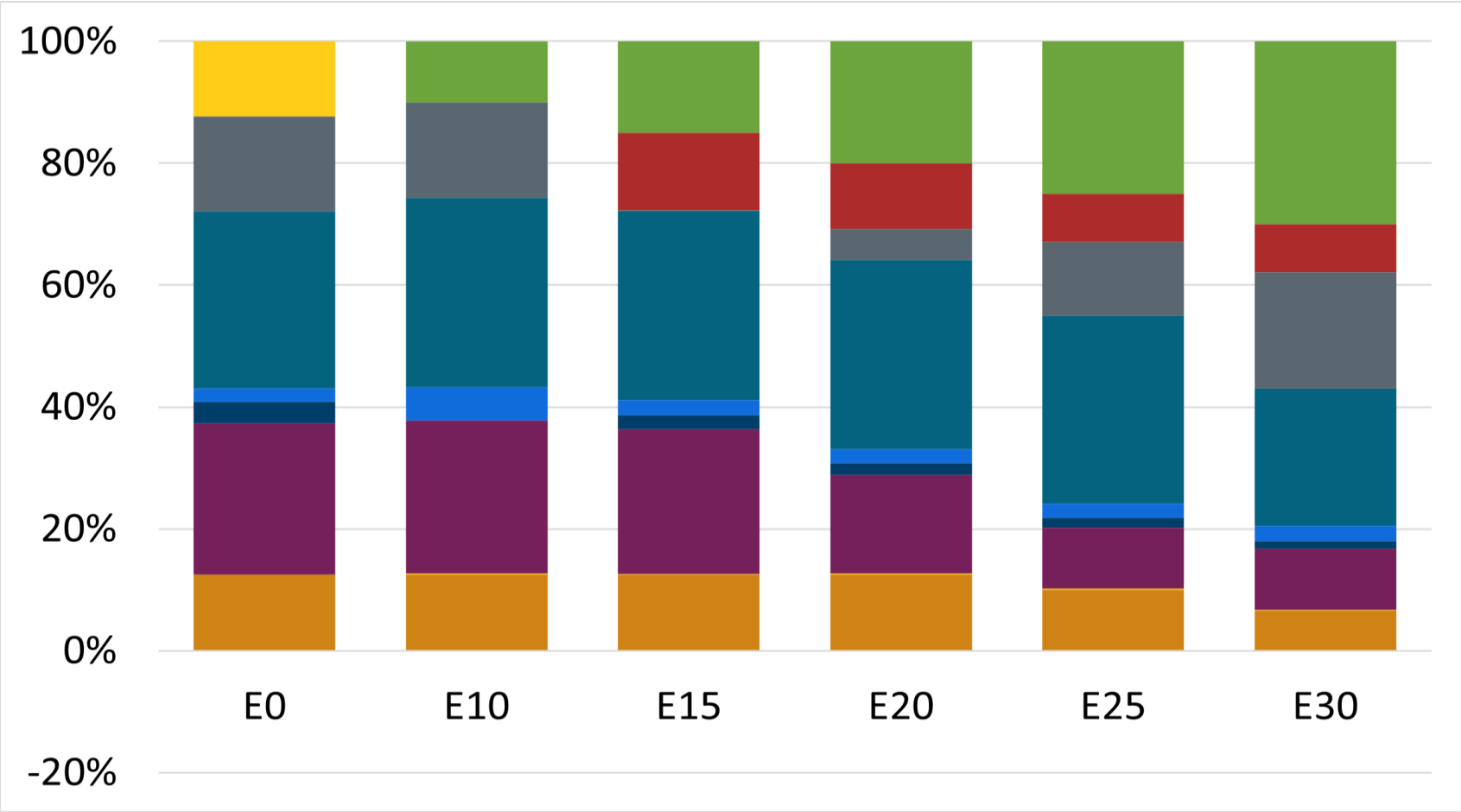
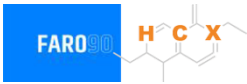
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US\$/gal)	\$ 2.423	\$ 2.395	\$ 2.331	\$ 2.266	\$ 2.201	\$ 2.134

Regular Gasoline ZM – Constant Octane



Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (US\$/gal)	\$ 2.280	\$ 2.265	\$ 2.199	\$ 2.131	\$ 2.061	\$ 1.990

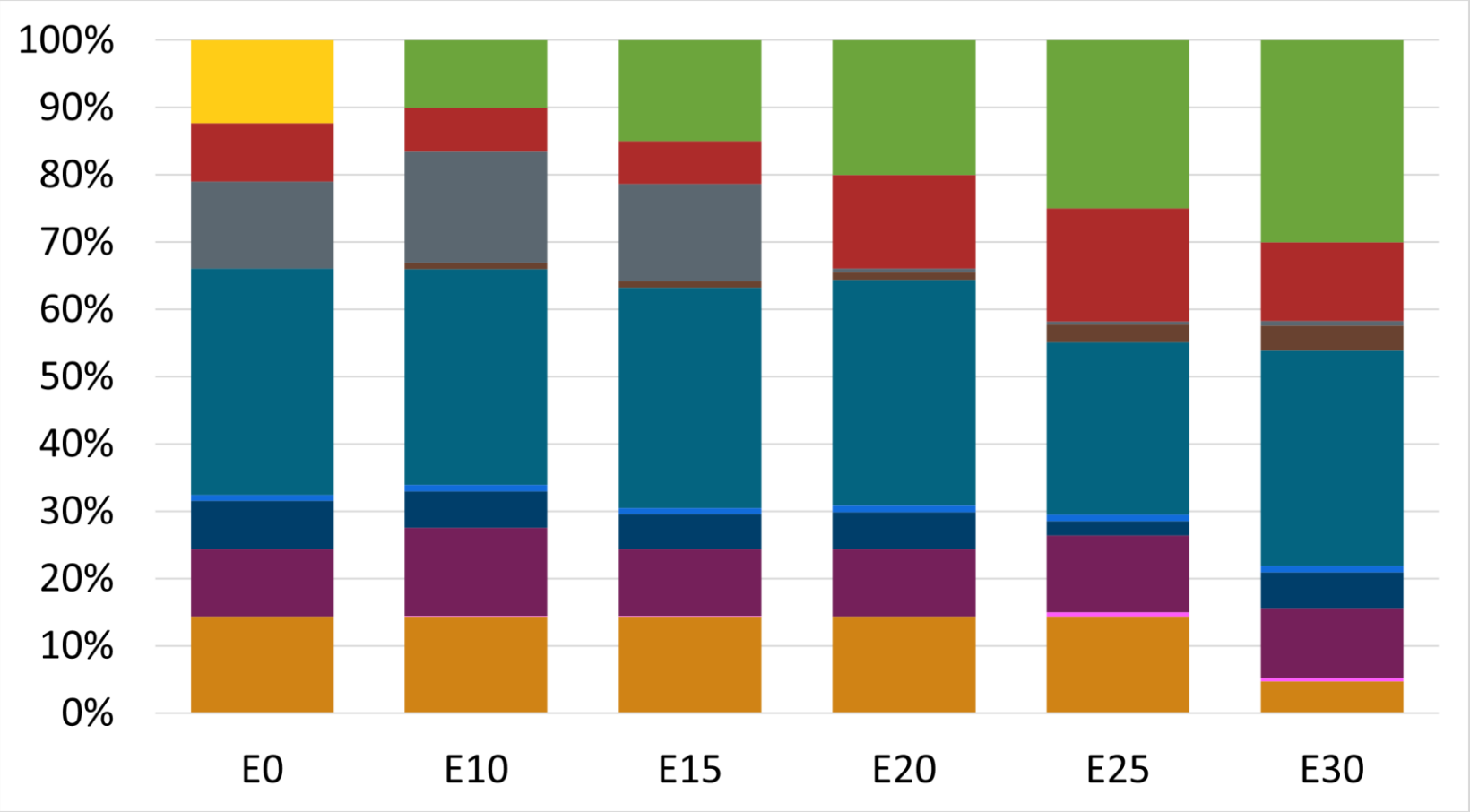
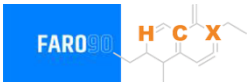
Premium Gasoline ZM – Constant Octane



Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US\$/gal)	\$ 2.478	\$ 2.442	\$ 2.377	\$ 2.310	\$ 2.242	\$ 2.172

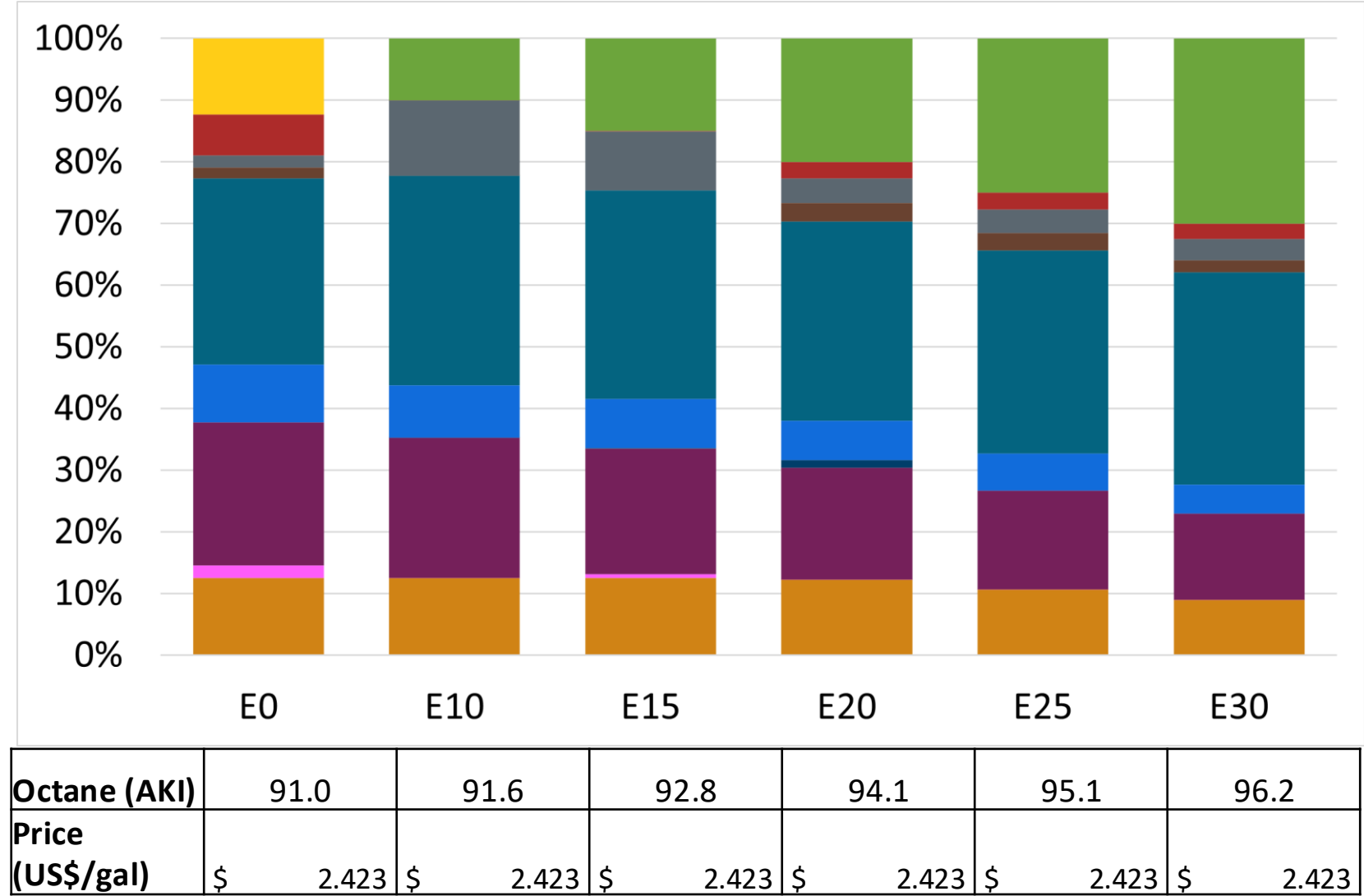
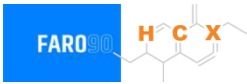


Regular Gasoline RP – Increased Octane

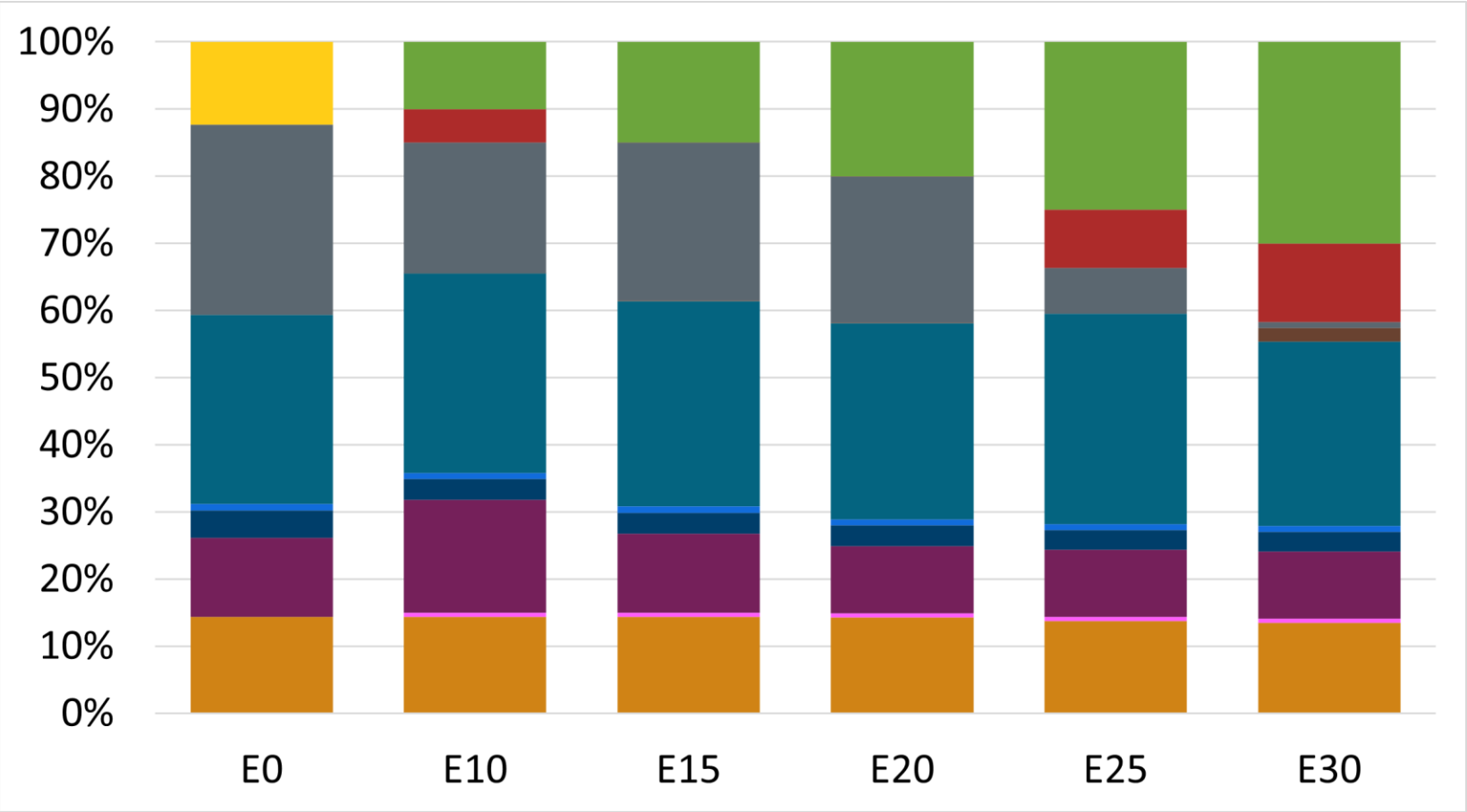


Octane (AKI)	87.0	87.2	88.6	90.1	90.5	92.9
Price (US\$/gal)	\$ 2.423	\$ 2.423	\$ 2.423	\$ 2.423	\$ 2.423	\$ 2.423

Premium Gasoline RP – Increased Octane

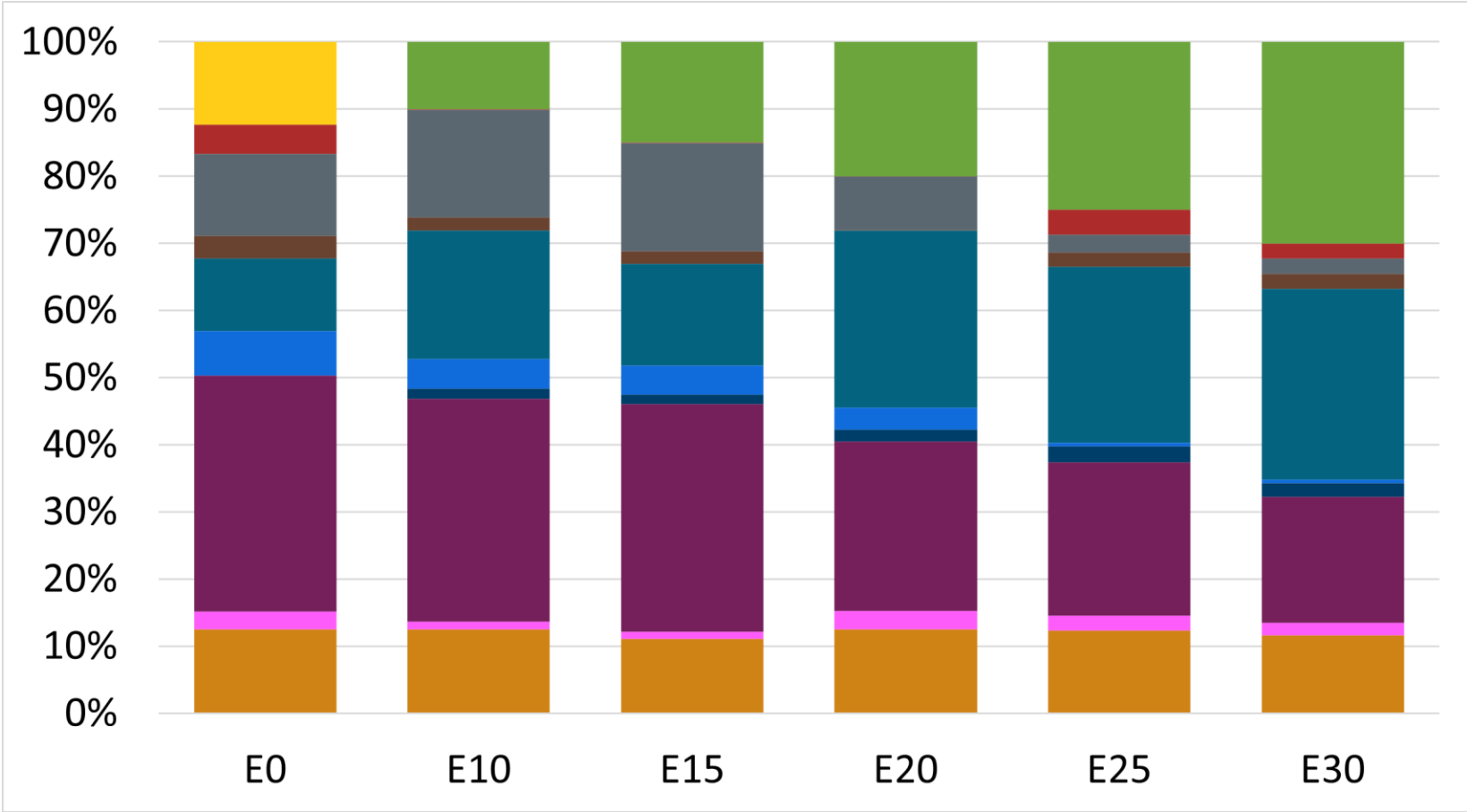


Regular Gasoline ZM – Increased Octane



Octane (AKI)	87.0	87.4	88.9	90.4	91.7	93.2
Price (US\$/gal)	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280	\$ 2.280

Premium Gasoline ZM – Increased Octane



Octane (AKI)	91.0	91.6	93.0	94.3	95.1	96.5
Price (US\$/gal)	\$ 2.475	\$ 2.475	\$ 2.475	\$ 2.475	\$ 2.475	\$ 2.475

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

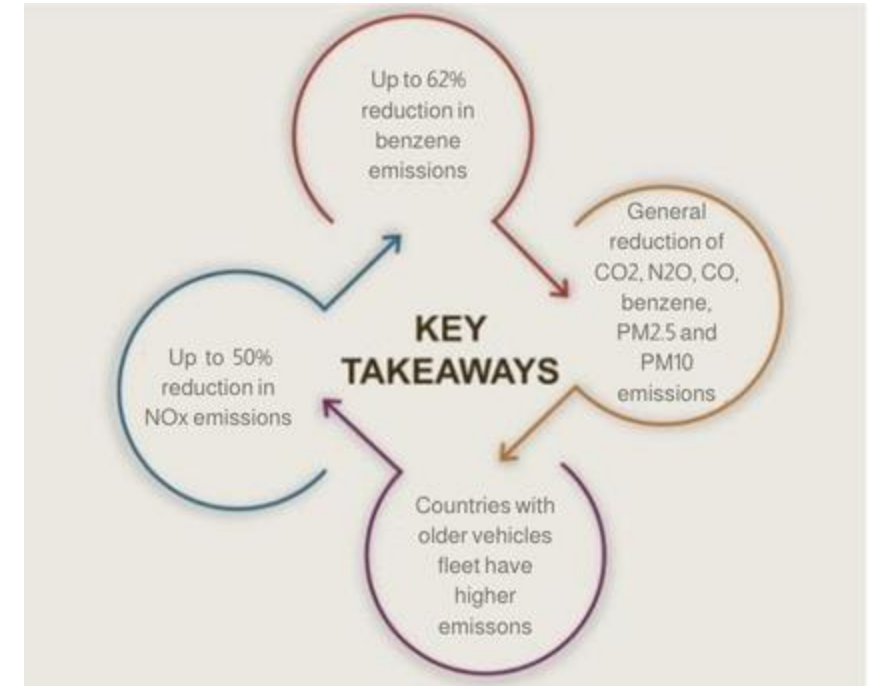
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

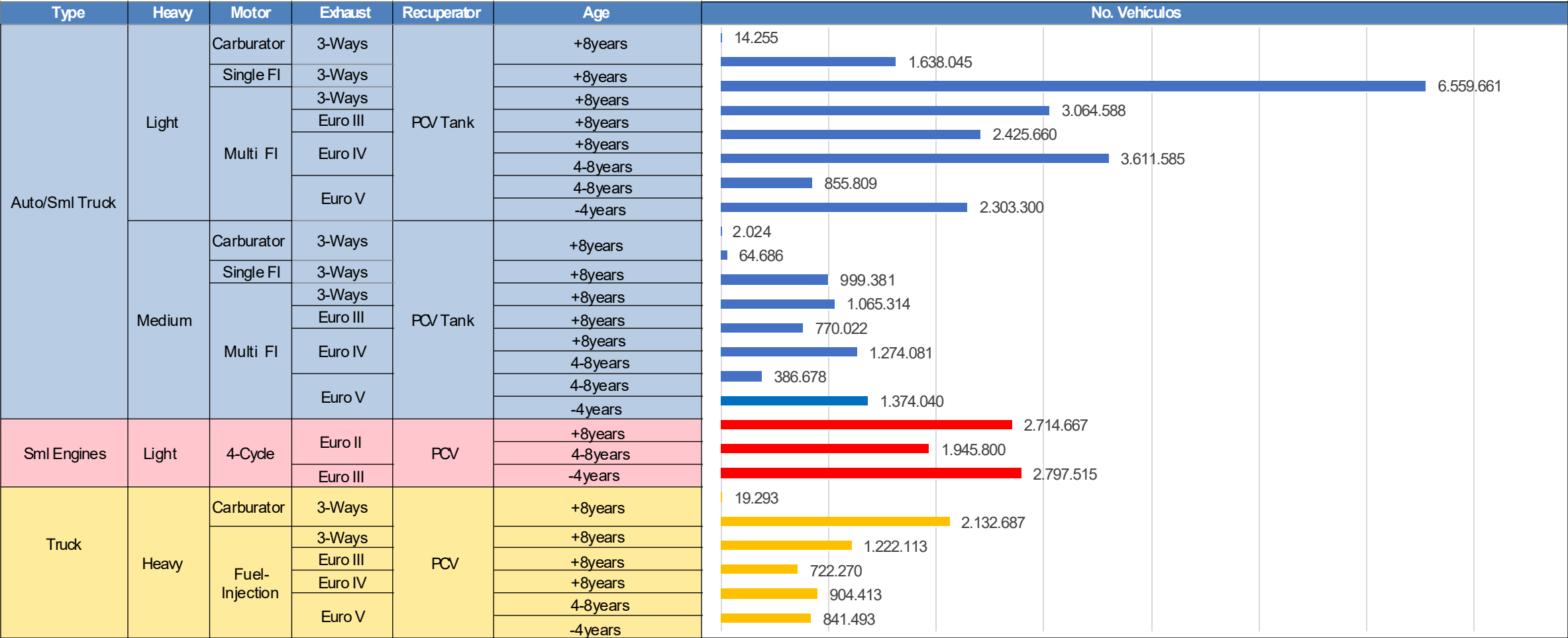
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Vehicular Fleet in Mexico



Gasoline Vehicular Fleet: **46,767,837**

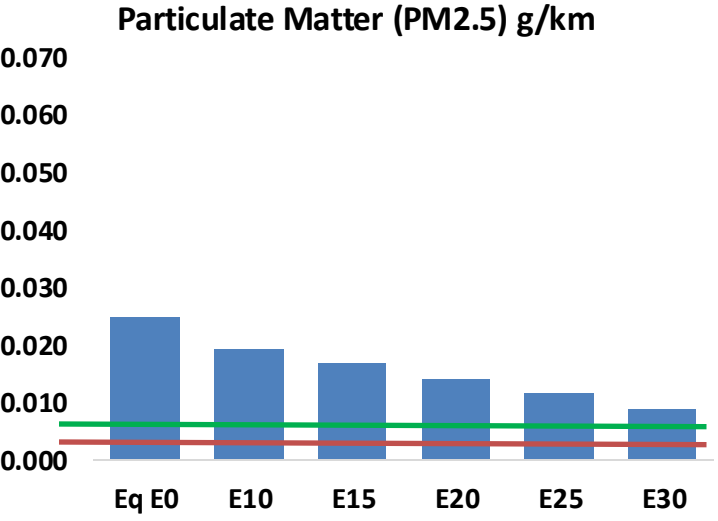
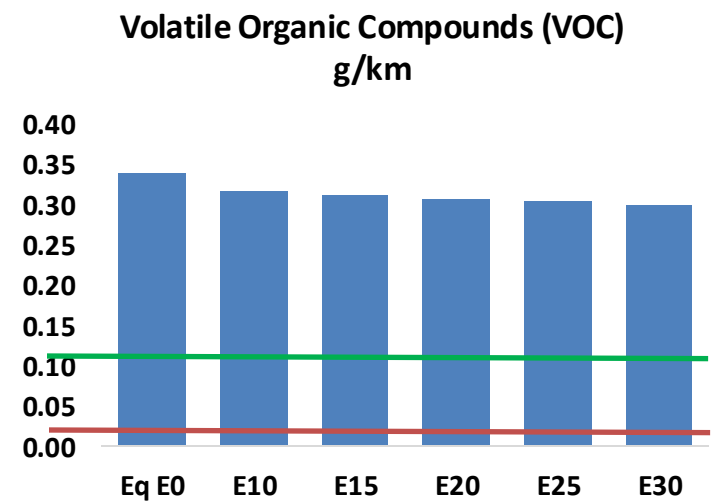
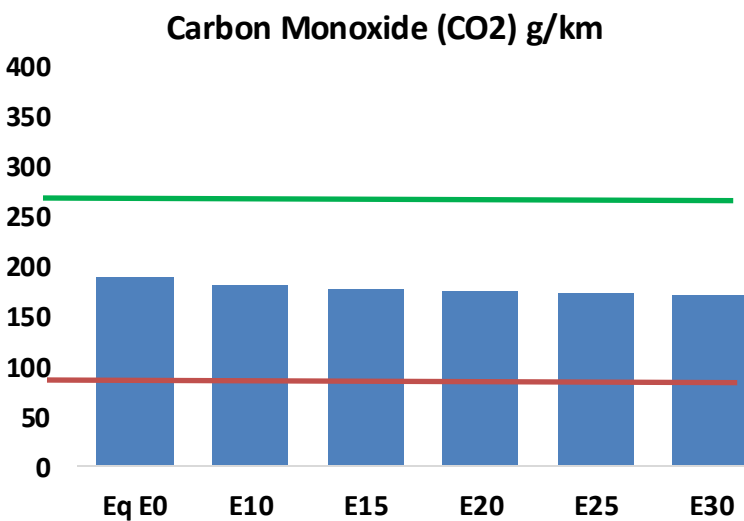
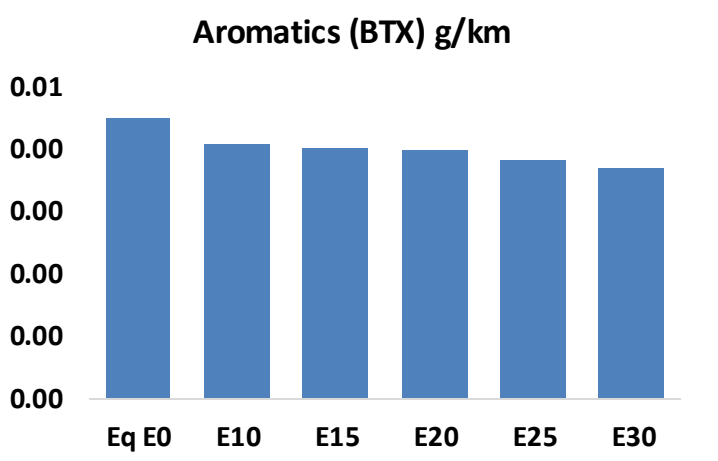
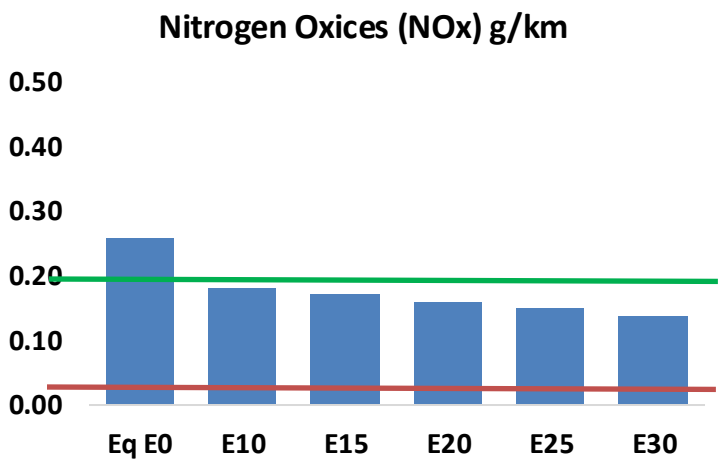
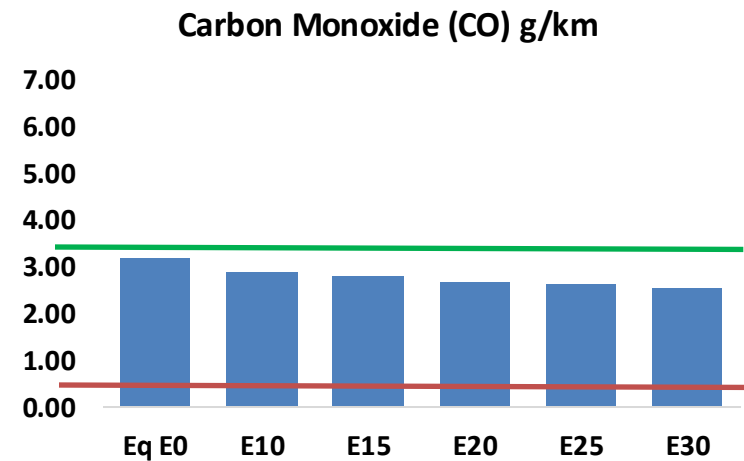
Average Age: **10 years**

Total = 60,978,233

Source: INEGI

Vehicular Emissions in Mexico

United States
Euro 6



Mexico – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	3,759,444	3,585,372	3,411,300	3,291,082	3,180,427	3,098,455	2,987,927	-9%	-15%
CO2	223,825,785	218,070,264	212,634,495	208,359,423	206,243,949	205,112,057	201,331,252	-5%	-8%
VOC	398,685	386,175	373,666	366,955	361,830	358,381	352,043	-6%	-9%
NOx	304,668	228,501	213,267	201,004	189,564	176,951	163,607	-30%	-38%
SOx	2,444	2,260	2,075	1,919	1,769	1,606	1,435	-15%	-28%
NH3	75,029	74,286	73,543	73,892	74,724	75,308	75,794	-2%	0%
N2O	3,288	3,270	3,257	3,273	3,341	3,378	3,416	-1%	2%
CH4	100,587	100,587	100,587	102,096	104,309	106,220	108,031	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,679	3,598	3,518	3,491	3,482	3,480	3,459	-4%	-5%
Acetaldehyde	8,132	9,333	13,694	20,854	28,412	32,466	38,362	68%	249%
Formaldehyde	26,777	26,033	30,348	35,635	37,333	41,300	44,960	13%	39%
Benzene	5,294	5,087	4,820	4,744	4,705	4,499	4,354	-9%	-11%
PM2.5	14,276	12,676	11,076	9,618	8,168	6,628	5,023	-22%	-43%
PM10	14,969	13,292	11,614	10,086	8,564	6,950	5,267	-22%	-43%

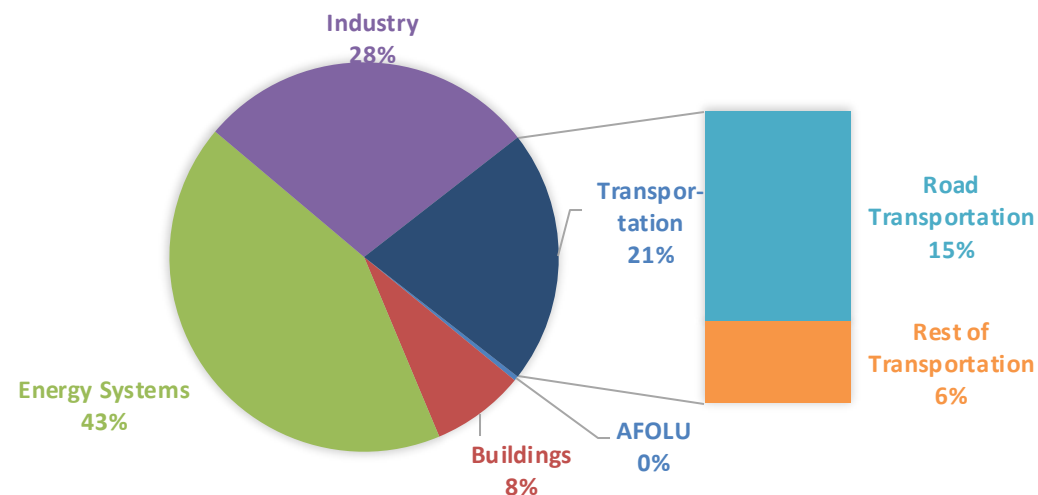
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

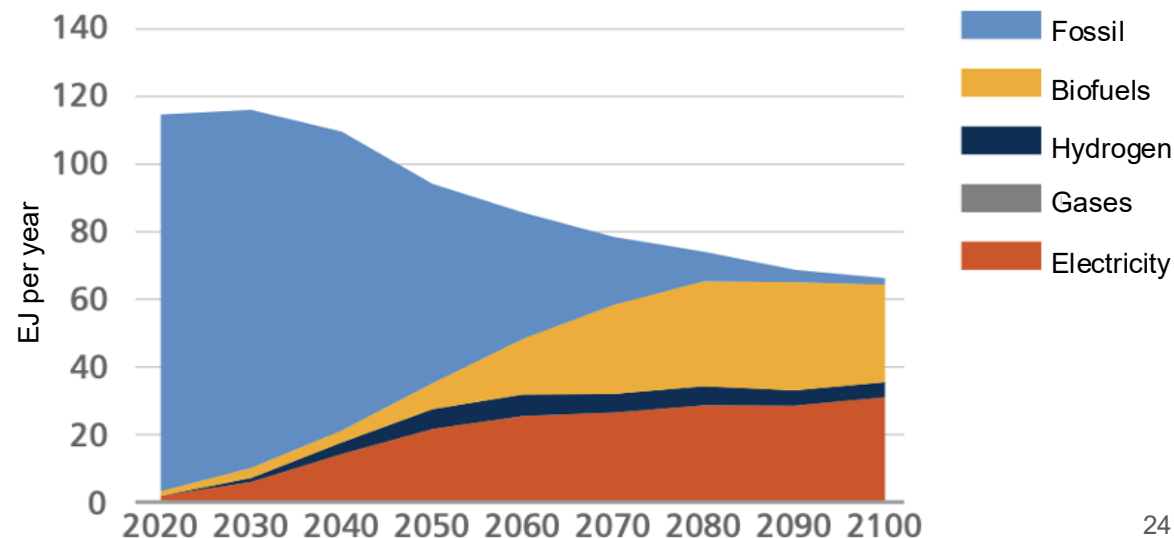
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

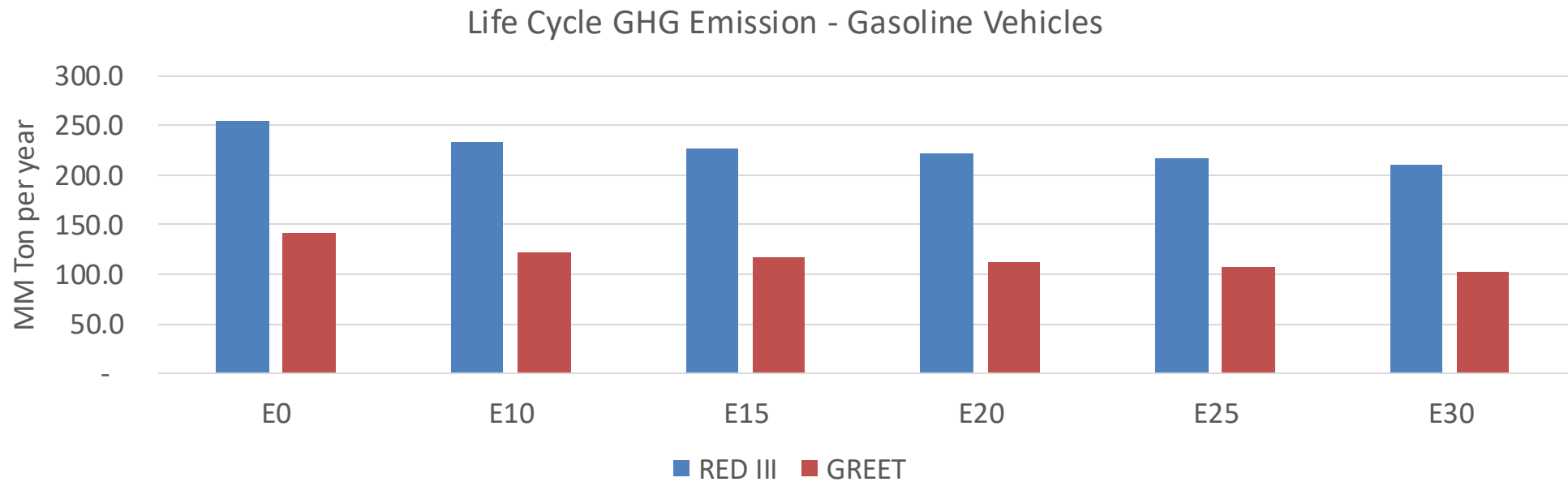
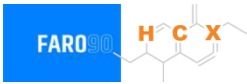
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Mexico – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	751.4
2035 Target (MMT/year)	500.4
Delta	251.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.4%	17.3%	20.6%	23.8%	27.8%
RED II/III	0.0%	8.1%	10.6%	12.7%	14.7%	17.1%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.6%	9.7%	11.6%	13.4%	15.7%
RED II/III	0.0%	8.2%	10.7%	12.8%	14.9%	17.3%